CSE 101 Project (Spring 2026)

Students Grade System

Implement a well-structured Python program that enables an instructor to maintain his students grades. The grade information is kept in a text-file of the form:

```
91007# Ahmad    Said#    50.0  78.5   73.2
91004#  Hassan  Khan#    45.5    36.7   88.5
91003#  Suleiman Wasim# 72.6  66.4   56.7
91002#  Majed  Sameer#     60.0  78.4   45.6
91006#  Muhammad  Adel# 85.5  69.8  44.5
91005#  Muhsim Zuheir#      70.0  62.1 95.4
91001#  Muneeb  Abdullatif#  30.0 56.5  44.8
```

The '#' symbol separates the id from the name and from the grades i.e. each line has 2 '#' symbols. The grades of each student are separated by one or more blanks. Each line of the text-file contains a **unique** student ID, the student first and last names, followed by test grades. No fixed number for number of test.

When your program starts, it will read all the information in the input file into list(s) or Dictionary. Then it will display the menu shown below.

1. Display Grade Info for all students
2. Display Grade Info for a particular student
3. Display tests average for all students
4. Modify a particular test grade for a particular student
5. Add test grades for a particular test for all students
6. Add a new Student
7. Delete a student
8. Save and Exit

Please select your choice:

Your program must loop as long as **option 8** has not been selected. It must display an appropriate error message if an invalid choice is entered. **Each of the options must be implemented in separate function.**

Once the user chooses option 8, your program will save the modified data in the same file that was used for reading at the beginning and with the same format. **Note that the saving to a file will be done only for option 8. All the work before will be in the list (dictionary).**

The options must have the following behaviors:

Option 1: Display Grade Info for all students

It displays the grade information of all students. It then waits for the Enter key to be pressed before returning control to the main menu:

Please select your choice: **1**

StudentID	Student Name	Test1	Test2	Test3
91007	Ahmad Said	50.0	78.5	73.2
91004	Hassan Khan	45.5	36.7	88.5
91003	Suleiman Wasim	72.6	66.4	56.7
91002	Majed Sameer	60.0	78.4	45.6
91006	Muhammad Adel	85.5	69.8	44.5
91005	Muhsim Zuheir	70.0	62.1	95.4
91001	Muneeb Abdullatif	30.0	56.5	44.8

Press Enter key to continue . . .

Note: The number of quizzes displayed depends on the number of quizzes in the grades text-file. Initially, if the file has no test grades; it has only student IDs and their corresponding names.

Option 2: Display Grade Info for a particular student

It prompts for and reads the studentID. It then searches for this studentID in the list or dictionary. If the ID is not found an appropriate error message is displayed, otherwise; the student information is displayed. In both cases, the option waits for the Enter key to be pressed before returning control to the main menu.

Please select your choice: **2**
Enter studentID: 9**1006**

StudentID	Student Name	Test1	Test2	Test3
91006	Muhammad Adel	85.5	69.8	44.5

Press Enter key to continue . . .

Please select your choice: **2**
Enter accountID: **91552**

Error: Invalid student ID
Press Enter key to continue . . .

Option 3: Display quiz averages for all students

The option displays the quiz grades and quiz averages of all students. Control is then returned to the main menu after pressing the Enter key.

```
Please select your choice:  3

StudentID      Student Name          Average

    91007      Ahmad    Said           67.2
    91004      Hassan  Khan            56.9
    91003      Suleiman Wasim         65.2
    91002      Majed  Sameer          61.3
    91006      Muhammad  Adel         66.6
    91005      Muhsim Zuheir          75.8
    91001      Muneeb  Abdullatif     43.8

Press Enter key to continue . . .
```

Note: The average depends on the number of tests. Your program must work correctly for any number of tests taken. An appropriate message must be displayed if students have not taken any test.

Option 4. Modify a particular test grade for a particular student

It prompts for and reads a studentID, the test number, and the new test grade. If the test number is invalid, or the new test grade is negative or greater than 100 an appropriate error message is displayed; otherwise it searches for this studentID. If the studentID is not found an appropriate error message is displayed, otherwise; the student grade is updated in the list.

```
Please select your choice:  4
Please enter studentID: 91003
Please enter quiz number to modify:  2
Please enter new quiz 2 grade:  80.0

Before grade modification: 91003   Suleiman Wasim    72.6    75.0    85.0    90.0    80.7
After grade modification: 91003   Suleiman Wasim     72.6    80.0    85.0    90.0    80.7

Press Enter key to continue . . .
```

```
Please select your choice:  4
Please enter quiz number to modify:  6
Error: Invalid quiz  number

Press Enter key to continue . . .
```

Control is returned to the main menu after pressing the Enter key.

Option 5: Add test grades for the next test for all students

It prompts the user to enter student grades for test **n + 1**, where **n** is the current number of tests for each student. The program then prompts for and reads the new test grade of each student and update the test list or dictionary of each student.

```
Please select your choice:  5
Please enter test grades for Test#4

Please enter grade for student : 91007
60.5
Please enter grade for student : 91004
70.0
Please enter grade for student : 91003
85.0
Please enter grade for student : 91002
55.5
Please enter grade for student : 91006
95.0
Please enter grade for student : 91005
90.5
Please enter grade for student : 91001
66.5

Press Enter key to continue . . .
```

Note: An error message must be displayed when any invalid grade is entered.

Option 6: Add New Student

It prompts for and read the ID of the student to be added. It will check if a student with same id already exists in the Students list. If not, it will be added by reading the remaining information i.e. name and quizzes and added as a student object to the array of students.
If the student with same id already exists, an error message will be displayed.

Option 7: Delete Student

To implement option 7, search the Students list r dictionary for the studentID of the student to be deleted. If found, delete it from the list/dictionary. If the studentID does not exist, display an error;

Option 8: Save and Exit

Save all data to the file that was used for reading at the beginning of the program, then terminates the program.

The following items must be observed when you write your code:

- Comments are important they are worth. (worth 5%)
- The code must use meaningful variable names and modular programming (worth 10%)
- Global variables are not allowed. You should learn how to pass parameters to functions and receive results.
- You must submit a working program. Non-working parts can be submitted separately. If a team submits a non-working program, it loses 20% of the grade.
- User input must be validated by the program i.e. valid range and valid type
- Your code has to be limited to the material covered in the lectures.
- The deadline for submitting the project is April 21 before midnight.

Deliverable:

Each team has to submit:

- The source code
- The report as part of the Jupyter notebook or as a separate word file. The report will describe how you solved the problem. In addition, you need to describe the different functions with their task and screen shots of your running code. (worth 5%)

Demo/Presentation:

- The period of **April 26 to May 02** will be used for project presentations.
- A slot of 15-20 minutes will be allocated to each team for their presentation and questions
- Students who do not appear for demo/presentation **will get 0**.

Note:

- **The project must be your own work.**
- **You must not share code with any other project group.**
- **Teams of 2 students will be formed.**
- **The project will be submitted on Canvas. The deadline is indicated above.**
- **Cheating in any form will result in a grade of F in the course project work.**